

LESSON 10.3

EXPLORING INEQUALITIES – PART 2



LESSON INTRODUCTION

MA.7.AR.2.1 Write and solve one-step inequalities in one variable within a mathematical context and represent solutions algebraically or graphically.

LEARNING TARGETS

- I can determine the minimum or maximum value that makes an inequality true.
- I can relate the solution set of an inequality to the context it represents.
- I can explore the relationship between the coefficients of the terms in an inequality and solution set.

BENCHMARK COHERENCE

BUILDING ON

MA.6.AR.1.2 Translate a real-world written description into an algebraic inequality in the form of $x > a$, $x < a$, $x \geq a$, or $x \leq a$. Represent the inequality on a number line.

MA.6.AR.2.1 Given an equation or inequality and a specified set of integer values, determine which values make the equation or inequality true or false.

WORKING TOWARD

MA.8.AR.2.2 Solve two-step linear inequalities in one variable and represent solutions algebraically and graphically.

ESSENTIAL QUESTIONS

- What is the minimum or maximum value that makes the inequality true?
- How do you relate the solution set of an inequality to the context it represents?
- How do you determine if solutions are reasonable for the context?
- What is the effect multiplying and dividing by a negative number has on the inequality?

LESSON NARRATIVE

In this lesson, students read, graph, and interpret inequalities in the form $px < q$ and $xp < q$. This lesson is an exploration with scaffolding where students connect algebraic representations of an inequality to a number line and a real-world context. Students explore the effect multiplying and dividing an inequality has on the inequality. Students explore the meaning of a minimum and maximum value in terms of inequalities and connect these to a real-world context.

COMPONENT SUMMARY

10.3.1 WARM-UP (~5 MIN.)

Write and interpret inequalities based on hanger diagrams

10.3.3 BRING IT TOGETHER (5 MIN.)

Relate the solution set to the context it represents

10.3.5 WRAP-UP (~5 MIN.)

Conjecture about the relationship between inequality symbol and sign of the coefficient

10.3.2 EXPLORATION (15 MIN.)

Explore determining maximum and minimum values of inequalities

10.3.4 EXPLORATION (15 MIN.)

Explore the effect multiplying or dividing by a negative number has on the inequality symbol

RESOURCES

GLOSSARY

Key Terms:

- Mathematical models

Additional Terms:

- Inequality
- Boundary points
- Minimum
- Maximum
- Conjecture

MATERIALS

- Calculators
- (Optional) Chart paper & markers

ADDITIONAL PREPARATION

- Consider editing the real-world problems based on the interests of your students.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may struggle to read and interpret inequalities with the variable on the right side.
- Students may struggle interpreting the solution set in the context of the problem.

THINGS TO CONSIDER

- Consider creating and displaying an anchor chart with vocabulary definitions, instructions, examples, and visuals.
- Consider asking students to practice reading inequalities from right to left and left to right.

LITERACY AND LANGUAGE ROUTINES

Notice and Wonder

In 10.3.4 Exploration, ask students to discuss what they notice and wonder about what happens when they multiply or divide an inequality by a negative. Help consolidate student understanding with a whole-class discussion.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.6.1 Reasonableness

In 10.3.2 Exploration, when determining the solution set for an inequality, ask students to consider if the solution makes sense in the context they represent. Students discover that within a real-world context, often a negative value does not make sense.

DIFFERENTIATE INSTRUCTION

The number lines used in 10.3.2 and 10.3.4 Explorations provide visual entry points. The partner and whole-class discussions provide auditory entry points.

10.3.1 WARM-UP: VISUAL MODELS

TEACHER MOVES

As students complete the Warm-Up, circulate the room, listening for student misconceptions. Consider creating a chart to make notes that may inform remediation.

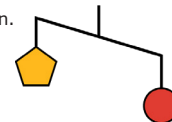
ENRICHMENT

Is there another way you could write the inequality $p < c$? Can you read the inequality from right to left?

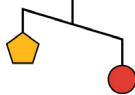
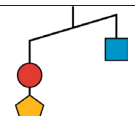
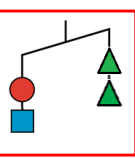


10.3.1 Warm-Up: Visual Models

An unbalanced hanger diagram is shown. It shows that the weight of one circle is greater than the weight of one pentagon.



1. Complete the table.

Hanger Diagram	Description	Inequality
	The weight of one circle, c , is greater than the weight of one pentagon, p .	$p < c$
	<i>The weight of one pentagon, p, is greater than the weight of one square, s.</i>	$s < p$
	The weight of one square, s , is <i>less</i> than the weight of a circle and a pentagon.	$c + p > s$
	<i>The weight of one square, s, and one circle, c, together is greater than the weight of two triangles, t.</i>	$s + c > 2t$



10.3.2 Exploration: Determining the Minimum or Maximum Value that Makes an Inequality True

Work with your partner to complete the following.

1. Bak Middle School of the Arts in West Palm Beach, Florida is hosting a rehearsal for its upcoming musical. The stage manager has \$83 from a recent fundraiser that can be used to order lunch for the cast and crew. The school's cafeteria can make boxed lunches for a cost of \$6 each. The inequality $6n \leq 83$ represents the situation, where n represents the number of boxed lunches that can be ordered from the cafeteria.
 - a. Select the values that represent a valid number of boxed lunches the stage manager can order from the cafeteria.

☐ 14	☐ 9.5
☐ $13\frac{5}{6}$	☒ 2
☒ 12	☐ -4
 - b. Explain why some values in the previous question are not valid in this context.
14 is too large of a number and -4 does not make sense in this context.
 - c. The mathematical solution to the inequality $6n \leq 83$ is $n \leq 13\frac{5}{6}$. What is the maximum number of boxed lunches the stage manager can order from the cafeteria?
13 is the maximum number because the cafeteria may not sell $\frac{5}{6}$ a lunch.
 - d. What is the minimum number of boxed lunches the stage manager can order from the cafeteria?
The least the stage manager can order is 0 boxed lunches.
 - e. The graph of the solution set for $6n \leq 83$ is shown.



10.3.2 EXPLORATION: DETERMINING THE MINIMUM OR MAXIMUM VALUE THAT MAKES AN INEQUALITY TRUE

QUESTIONING

Remind students that they can select multiple values for question 1a.

TEACHER MOVES

Encourage students to substitute each value into the inequality to test if it is true. For question 1b, consider highlighting the fact that -4 would not make sense in the context of the problem.

ENRICHMENT

- What is the difference between maximum and minimum?
- How can you determine the maximum in this situation?
- How can you determine the minimum?
- For question 1c, should you round up or down? Why?

10.3.2 EXPLORATION: DETERMINING THE MINIMUM OR MAXIMUM VALUE THAT MAKES AN INEQUALITY TRUE (CONTINUED)

DIFFERENTIATE INSTRUCTION

Consider providing students with the following fill-in-the-blank number sentence:

_____ $\times$ _____ = _____.

ENRICHMENT

- If the maximum distance Hiromi can walk is 2.775 miles, how can we write that using an inequality?
- If 2.775 is the maximum, which way should the inequality symbol be facing?
- Should 2.775 be included or not included in the solution?

TEACHER MOVES

Think-Pair-Share

For question 2c, have students think independently first before sharing with a partner. Ask students to consider whether negative numbers are possible in this context.

Discuss with your partner how the graph that mathematically represents the solution set is not the best representation of the real-world context. Then, represent the solutions in the real-world context on the number line shown.

This is not a good representation because you cannot buy a negative number of lunch boxes.



2. Hiromi walks around his neighborhood each afternoon, tracking his pace each day. The fastest rate Hiromi can walk for an extended period of time is 3.7 miles per hour.

- a. What is the maximum distance Hiromi can walk in 0.75 hours?

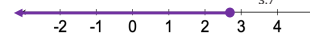
2.775 miles

- b. The inequality $\frac{d}{3.7} \leq 0.75$ can be used to represent the distance Hiromi walks, d , in 0.75 hours given his fastest walking rate of 3.7 miles per hour.

Use your answer to Part A to determine the solution set for $\frac{d}{3.7} \leq 0.75$.

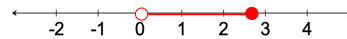
$d \leq 2.775$ miles

- c. The graph of the solution set for $\frac{d}{3.7} \leq 0.75$ is shown.



Discuss with your partner how the graph that mathematically represents the solution set is not the best representation of the real-world context. Then, represent the solutions in the real-world context on the number line.

This is not a good representation because you cannot run negative miles.





10.3.3 Bring It Together: Solutions and Mathematical Models in Context

Each of the inequalities in the Exploration has a mathematical solution set that can be represented on a number line. These graphs are examples of mathematical models.



Mathematical models are helpful tools used to understand a situation and make predictions. However, sometimes the mathematical model can include parts that do not make sense in a real-world context. When interpreting models to solve problems, it is important to analyze whether aspects of the model make sense in a given context.

10.3.3 BRING IT TOGETHER: SOLUTIONS AND MATHEMATICAL MODELS IN CONTEXT

TEACHER MOVES

Consider utilizing some time in this section to address misconceptions that students had during the previous 10.3.2 Exploration.



10.3.4 Exploration: Matching Inequalities to Solutions

1. Complete the tables by substituting each value of x to determine if it is a solution to each inequality.

Value of x	What is the value of $3x$?	Is $3x \geq -15$ true or false?	Solution?
-10	-30	False	☐ Yes ☐ No
-7.4	-22.2	False	☐ Yes ☐ No
$2\frac{1}{8}$	$6\frac{3}{8}$	True	☐ Yes ☐ No
6	18	True	☐ Yes ☐ No

Value of x	What is the value of $-3x$?	Is $-3x \geq 15$ true or false?	Solution?
-10	30	True	☐ Yes ☐ No
-7.4	22.2	True	☐ Yes ☐ No
$2\frac{1}{8}$	$-6\frac{3}{8}$	False	☐ Yes ☐ No
6	-18	False	☐ Yes ☐ No

2. The solution sets for the inequalities $3x \geq -15$ and $-3x \geq 15$ are given in the table. Write the inequality that matches each graph of the solution set.

Inequality	Graph of the Values of x
$3x \geq -15$ $x \geq -5$	
$-3x \geq 15$ $x \leq -5$	

10.3.4 EXPLORATION: MATCHING INEQUALITIES TO SOLUTIONS

DIFFERENTIATE INSTRUCTION

The number lines in question 2 provide students with visual entry points. The partner and whole-class discussions provide them with an auditory entry point.

ENRICHMENT

Notice and Wonder

- What do you notice? What is the same? What is different?
- What does that make you wonder or think about these inequalities?

10.3.4 EXPLORATION: MATCHING INEQUALITIES TO SOLUTIONS (CONTINUED)

TEACHER MOVES

If time is running short, consider assigning the remaining questions in the 10.3.4 Exploration and 10.3.5 Wrap-Up for homework.

ENRICHMENT

Notice and Wonder

- What do you notice? What is the same? What is different?
- What does that make you wonder or think about these inequalities?

TEACHER MOVES

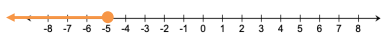
Consider having a whole-class discussion to consolidate student understanding regarding what to do when multiplying or dividing by a negative number.

3. Complete the tables by substituting each value of x to determine if it is a solution to each inequality.

Value of x	What is the value of $\frac{x}{-1}$?	Is $\frac{x}{-1} \leq 5$ true or false?	Solution?
-8.2	8.2	False	☐ Yes ☐ No
-6	6	False	☐ Yes ☐ No
0	0	True	☐ Yes ☐ No
$\frac{14}{3}$	$-\frac{14}{3}$	True	☐ Yes ☐ No

Value of x	What is the value of $\frac{x}{2}$?	Is $\frac{x}{2} \leq -2.5$ true or false?	Solution?
-10	-5	True	☐ Yes ☐ No
-7.4	-3.7	True	☐ Yes ☐ No
$2\frac{1}{8}$	$1\frac{1}{16}$	False	☐ Yes ☐ No
6	3	False	☐ Yes ☐ No

4. The solution sets for the inequalities $\frac{x}{-1} \leq 5$ and $\frac{x}{2} \leq -2.5$ are given in the table. Write the inequality that matches each graph of the solution set.

Inequality	Graph of the Values of x
$\frac{x}{-1} \leq 5$ $x \geq -5$	
$\frac{x}{2} \leq -2.5$ $x \leq -5$	

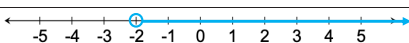
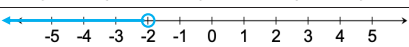
5. Look back at your answers to questions 2 and 4. What do you notice about the original inequality symbol and the direction of the arrows on the graphs of the solution sets?

Answers will vary. I notice that the original inequality symbol flipped and so did the graphs arrows when I multiplied or divided by a negative number.

6. Compare your work on questions 1-5 with your partner. Discuss any differences and revise your work if necessary.

Answers will vary.

7. The graphs of two solution sets are shown.

Graph A	
Graph B	

The solution set for each inequality in the table matches one of the graphs. Based on your work and observations thus far, work with your partner to determine which graph matches each inequality.

Inequality	Which Graph?
$-2p > 4$	Graph B
$3.8p > -7.6$	Graph A
$\frac{p}{3} > -\frac{2}{3}$	Graph A
$-\frac{p}{5} > 0.4$	Graph B

8. Explain how you and your partner determined which graphs match each inequality.
The inequalities that required either division or multiplication using a negative number required the inequality symbol to be flipped. In both cases it led to the p being less than -2 .

**10.3.5 Wrap-Up: Cool Down**

1. Make a conjecture about the relationship between the inequality symbol and the sign of the coefficient of the variable term when determining the solution set for an inequality. Use examples from the lesson if they're helpful.

If the sign of the coefficient of the variable term is negative, then the inequality symbol will be flipped in the opposite direction.

10.3.5 WRAP-UP: COOL DOWN**DIFFERENTIATE INSTRUCTION**

Consider allowing students to use an audio recorder or speech-to-text software if they struggle expressing their response in writing. Students could also draw a diagram and provide examples to demonstrate mastery of the learning target.